Midterm Review: Chapters 1–2

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The midterm (Thu. Nov. 5) covers C&B Chapter 1 and §§2.1–2.3: the lecture notes for Weeks 1–5, and Homeworks 1–3. **Differentiating under an integral sign (§2.4) is not on the midterm.** The exam is closed book; you may bring one page of notes (a cheat sheet) and a calculator.

Part I. Probability Theory (Chapter 1, Weeks 1–3)

1. Sets and events (§1.1)

Key definitions.

- **Sample space** S: the set of all possible outcomes (Definition 1.1.1). An **event** is a subset of S (Definition 1.1.2).
- **Set operations:** union $A \cup B$, intersection $A \cap B$, complement A^c; commutative, associative, and distributive laws; **DeMorgan's laws** $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$ (Theorem 1.1.4). They extend to finite and countable collections.
- **Countable unions and intersections:** $x \in \bigcup_n A_n$ if $x \in A_n$ **for some** n; $x \in \bigcap_n A_n$ if $x \in A_n$ **for all** n.
- **Disjoint** (mutually exclusive): $A \cap B = \emptyset$. **Partition:** pairwise disjoint sets whose union is S (Definitions 1.1.5–1.1.6).

2. Sigma algebras and the axioms (§1.2.1)

Key definitions.

- **Sigma algebra** $\mathcal{B}$ (Definition 1.2.1): contains $\emptyset$; closed under complements; closed under countable unions. Consequences: $S \in \mathcal{B}$, and $\mathcal{B}$ is closed under countable intersections (by DeMorgan).
- **Kolmogorov axioms** (Definition 1.2.4): $P(A) \geq 0$; $P(S) = 1$; if $A_1, A_2, \ldots$ are pairwise disjoint, $P\big(\bigcup_i A_i\big) = \sum_i P(A_i)$.
- **Finite** S (Theorem 1.2.6): assigning $p_i \geq 0$ with $\sum p_i = 1$ to the outcomes and adding defines a probability function.

3. The calculus of probabilities (§1.2.2)

Key results.

- $P(\emptyset) = 0$, $P(A) \le 1$, $P(A^c) = 1 - P(A)$ (Theorem 1.2.8).
- $P(B \cap A^c) = P(B) - P(A \cap B)$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; if $A \subset B$ then $P(A) \le P(B)$ (Theorem 1.2.9).
- **Partition formula:** $P(A) = \sum_i P(A \cap C_i)$ for a partition $C_1, C_2, \ldots$ (Theorem 1.2.11(a)).
- **Boole's inequality:** $P\big(\bigcup_i A_i\big) \le \sum_i P(A_i)$ (Theorem 1.2.11(b)). **Bonferroni's inequality:** $P(A \cap B) \ge P(A) + P(B) - 1$ (1.2.9), and its n-event form (1.2.10).

4. Counting and equally likely outcomes (§1.2.3)

Key results.

- **Fundamental theorem of counting** (Theorem 1.2.14): a job of k tasks done in $n_1, \ldots, n_k$ ways can be done in $n_1 \times \cdots \times n_k$ ways.
- **Table 1.2.1** (choose r from n):

	Without replacement	**With replacement**
Ordered	$\dfrac{n!}{(n-r)!}$	n^r
Unordered	$\dbinom{n}{r}$	$\dbinom{n+r-1}{r}$

- **Multinomial coefficient:** $\dfrac{k!}{k_1! \cdots k_m!}$ arrangements of k objects with k_j identical objects of type j.
- **Equally likely outcomes:** $P(A) = N(A)/N(S)$.

5. Conditional probability and Bayes' rule (§1.3)

Key definitions and results.

- **Conditional probability** (Definition 1.3.2): $P(A \mid B) = P(A \cap B)/P(B)$ for $P(B) > 0$. For fixed B, $P(\cdot \mid B)$ satisfies the axioms, so every Week 1 rule holds conditionally, e.g. $P(A^c \mid B) = 1 - P(A \mid B)$.
- **Multiplication (chain) rule:** $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$.
- **Law of total probability:** $P(B) = \sum_i P(B \mid A_i)P(A_i)$ for a partition $A_1, A_2, \ldots$
- **Bayes' rule** (Theorem 1.3.5):

$$P(A_i \mid B) = \frac{P(B \mid A_i)\,P(A_i)}{\sum_j P(B \mid A_j)\,P(A_j)}.$$

Prior $P(A_i)$, likelihood $P(B \mid A_i)$, posterior $P(A_i \mid B)$.

6. Independence (§1.3)

Key definitions and results.

- A and B are **independent** if $P(A \cap B) = P(A)P(B)$ (Definition 1.3.7). Then A and B^c, A^c and B, and A^c and B^c are also independent (Theorem 1.3.9).
- **Mutual independence** of $A_1, \ldots, A_n$ (Definition 1.3.12): the product rule holds for **every** subcollection.

7. Random variables and cdfs (§§1.4–1.5)

Key definitions and results.

- A **random variable** is a function $X : S \to \mathbb{R}$ (Definition 1.4.1). It induces a probability on its range: $P(X \in A) = P\big(\{s \in S : X(s) \in A\}\big)$.
- **Cdf:** $F_X(x) = P(X \le x)$ for **all** real x (Definition 1.5.1).
- **Theorem 1.5.3:** F is a cdf if and only if (a) $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to \infty} F(x) = 1$; (b) F is nondecreasing; (c) F is right-continuous.
- **Continuous** if F_X is continuous; **discrete** if F_X is a step function (Definition 1.5.7). A cdf can be **neither** (a mix of jumps and continuous increase).
- **Identically distributed** (Definition 1.5.8) if and only if $F_X = F_Y$ everywhere (Theorem 1.5.10). Identically distributed does not mean $X = Y$.
- **Reading probabilities from a cdf:**

  $$P(a < X \le b) = F_X(b) - F_X(a), \qquad P(X = a) = F_X(a) - F_X(a^-), \qquad F_X(a^-) = \lim_{x \uparrow a} F_X(x).$$

8. Pmfs and pdfs (§1.6)

Key definitions and results.

- **Pmf:** $f_X(x) = P(X = x)$ (Definition 1.6.1). **Pdf:** $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt$ for all x (Definition 1.6.3), so $f_X = F_X'$ where f_X is continuous.
- **Theorem 1.6.5:** f_X is a pmf or pdf if and only if $f_X \ge 0$ and it sums or integrates to 1.
- For continuous X, $P(X = x) = 0$, so endpoints don't matter: $P(a < X < b) = P(a \le X \le b) = \int_a^b f_X(x)\,dx$.

Part II. Transformations and Expectations (Chapter 2, Weeks 4–5)

9. Distributions of functions of a random variable (§2.1)

Key definitions and results.

- **Supports** (2.1.7): $\mathcal{X} = \{x : f_X(x) > 0\}$ and $\mathcal{Y} = \{g(x) : x \in \mathcal{X}\}$.
- **Discrete X:** $f_Y(y) = \sum_{x \in g^{-1}(y)} f_X(x)$.
- **Cdf method** (2.1.4): $F_Y(y) = P(g(X) \le y) = \int_{\{x \in \mathcal{X} : g(x) \le y\}} f_X(x)\,dx$.
- **Monotone** g (Theorem 2.1.3): $F_Y(y) = F_X(g^{-1}(y))$ if g is increasing; $F_Y(y) = 1 - F_X(g^{-1}(y))$ if g is decreasing and X is continuous.
- **Change of variables** (Theorem 2.1.5): for monotone g,

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy} g^{-1}(y)\right|, \qquad y \in \mathcal{Y}.$$

- **Piecewise monotone** g (Theorem 2.1.8): split $\mathcal{X}$ into pieces $A_1, \dots, A_k$ on which g is monotone with the same image $\mathcal{Y}$, and add the contributions $f_X(g_i^{-1}(y))\left|\frac{d}{dy} g_i^{-1}(y)\right|$. Example: $Y = X^2$ gives $f_Y(y) = \frac{1}{2\sqrt{y}}\left[f_X(\sqrt{y}) + f_X(-\sqrt{y})\right]$ (Example 2.1.7).

10. The probability integral transformation (§2.1)

Key results.

- **Theorem 2.1.10:** if F_X is continuous, then $F_X(X) \sim$ uniform(0, 1).
- **Inverse cdf** (2.1.13): $F_X^{-1}(y) = \inf\{x : F_X(x) \ge y\}$. If $V \sim$ uniform(0, 1), then $F_X^{-1}(V)$ has cdf F_X.

11. Expected values (§2.2)

Key definitions and results.

- **Definition 2.2.1:** $\mathrm{E}\, g(X) = \int g(x) f_X(x)\,dx$ or $\sum g(x) f_X(x)$, **provided** $\mathrm{E}|g(X)| < \infty$. Otherwise the expectation does not exist (Cauchy, Example 2.2.4).
- **Linearity** (Theorem 2.2.5): $\mathrm{E}\big(a g_1(X) + b g_2(X) + c\big) = a\,\mathrm{E}\, g_1(X) + b\,\mathrm{E}\, g_2(X) + c$; expectation preserves inequalities.
- **But** $\mathrm{E}\, g(X) \ne g(\mathrm{E}X)$ in general.
- **Minimizing distance** (Example 2.2.6): $b = \mathrm{E}X$ minimizes $\mathrm{E}(X - b)^2$.

- **Tail formula** (Exercise 2.14, HW3): for nonnegative continuous X, $\mathrm{E}X = \int_0^\infty \left[1 - F_X(x)\right] dx$.

12. Moments and variance (§2.3)

Key definitions and results.

- **Moments:** $\mu_n' = \mathrm{E}X^n$; **central moments:** $\mu_n = \mathrm{E}(X-\mu)^n$ (Definition 2.3.1).
- **Variance** (Definition 2.3.2): $\mathrm{Var}\,X = \mathrm{E}(X - \mathrm{E}X)^2 = \mathrm{E}X^2 - (\mathrm{E}X)^2$.
- $\mathrm{Var}(aX+b) = a^2 \mathrm{Var}\,X$ (Theorem 2.3.4).
- **Examples to know:** exponential(λ): mean λ, variance λ^2; binomial(n, p): mean np, variance $np(1-p)$; Poisson(λ): mean and variance λ.

13. Moment generating functions (§2.3)

Key definitions and results.

- **Mgf** (Definition 2.3.6): $M_X(t) = \mathrm{E}e^{tX}$, which must be finite for all t in a **neighborhood of 0**, $-h < t < h$.
- **Moments from the mgf** (Theorem 2.3.7): $\mathrm{E}X^n = M_X^{(n)}(0)$.
- **Linear transformations** (Theorem 2.3.15): $M_{aX+b}(t) = e^{bt} M_X(at)$.
- **Mgfs to know:** binomial $[pe^t + (1-p)]^n$; Poisson $e^{\lambda(e^t-1)}$; gamma $(1-\beta t)^{-\alpha}$ for $t < 1/\beta$, with exponential(λ) $= (1-\lambda t)^{-1}$ for $t < 1/\lambda$.
- **Uniqueness** (Theorem 2.3.11): if two mgfs exist and agree near 0, the distributions are the same. Moments alone may **not** determine a distribution (Example 2.3.10, lognormal).
- **Convergence** (Theorem 2.3.12): if $M_{X_i}(t) \to M_X(t)$ near 0 and M_X is an mgf, then $F_{X_i}(x) \to F_X(x)$ at continuity points of F_X. With Lemma 2.3.14, $\left(1 + \frac{a_n}{n}\right)^n \to e^a$ when $a_n \to a$; this gives the Poisson approximation to the binomial (Example 2.3.13).
- **Existence hierarchy:** mgf exists $\Rightarrow$ all moments finite $\Rightarrow$ $\mathrm{Var}\,X < \infty \Rightarrow \mathrm{E}X$ exists, and no arrow reverses (Cauchy; $2/y^3$ on $y > 1$; lognormal).

Part III. Ways of Thinking

The topics change from week to week, but the same habits solve most of the problems. None of them is specific to probability: they work for any math or statistics problem, and for most problems outside the classroom too.

1. Always start from definitions. When you are stuck, write down the precise definition of every term in the question. The definition usually tells you what to check.

- *In this course:* "$A = B$" means $A \subset B$ and $B \subset A$. "Independent" means $P(A \cap B) = P(A)P(B)$. "Cdf" means $P(X \le x)$ for **every** real x. "E $g(X)$ exists" means $\mathrm{E}|g(X)| < \infty$.

- *Beyond:* before arguing about whether something is "fair," "significant," or "a success," agree on what the word means.

2. Build from a few rules, and justify every step. All of probability follows from three axioms. A proof is a chain of steps, each one justified by a definition, an axiom, or an earlier result.

- *In this course:* $P(A^c) = 1 - P(A)$, inclusion–exclusion, and Boole's inequality all come from the axioms. Cite the result you use at each step.
- *Beyond:* know which assumptions a conclusion rests on. If an assumption fails, so may the conclusion.

3. Translate words into precise objects. Name the events and random variables, write down what is given and what is asked, and only then compute.

- *In this course:* "given that" becomes a conditional probability; "at least one" becomes a union; a statement about X is a statement about an event $\{s : X(s) \in A\}$ in S.
- *Beyond:* most of the work in a real problem is stating it clearly. A well-posed question is half answered.

4. Draw a picture. Pictures show what to compute, even though they are not proofs.

- *In this course:* Venn diagrams for set identities; trees for conditional probabilities; a sketch of the cdf to see jumps and flat pieces; a graph of g on the support for transformations.
- *Beyond:* plot the data before modeling it.

5. Divide and conquer. Break a hard problem into pieces that do not overlap and together cover everything, solve each piece, then put them back together.

- *In this course:* write a set as a disjoint union and add (Axiom 3); the partition formula and total probability; counting task by task; piecewise-monotone transformations.
- *Beyond:* split a project into parts with clear boundaries, so nothing is done twice and nothing is missed.

6. Turn the problem around. If the direct route is hard, try the opposite question.

- *In this course:* the complement for "at least one"; Bayes' rule to turn $P(B \mid A)$ into $P(A \mid B)$; transform the *event*, not the density, in the cdf method; the inverse cdf to go from a uniform to any distribution.
- *Beyond:* "What would have to be true for this to fail?" is often easier to answer than "Why does this work?"

7. Reason about what you don't know, and update with evidence. When the answer depends on an unknown case, consider each case, weight it by its probability, and update the weights as information arrives.

- *In this course:* total probability and Bayes' rule; prior, likelihood, and posterior; the base-rate fallacy, where an accurate test still gives mostly false positives for a rare disease.

- *Beyond:* medical tests, spam filters, and court evidence all need the base rate, not only how accurate the evidence is.

8. Know where things live. Every formula comes with a domain, and every result comes with conditions.

- *In this course:* supports in transformations; the pieces of a cdf outside the support; $P(B) > 0$ for conditioning; for which t an mgf is finite; whether an expectation exists; the hypotheses of each theorem (continuity in Theorem 2.1.10, monotonicity in Theorem 2.1.5).
- *Beyond:* a model fit on one population may not apply to another.

9. Recognize patterns, and reuse what you know. Most new problems are old problems in disguise. Ask, “Have I seen something like this before?”

- *In this course:* recognize a known series or a pdf kernel inside a sum or integral; match an mgf to a known one, then use uniqueness; reduce to a standard case by a shift and scale.
- *Beyond:* build a toolbox of solved problems, and look for the one that fits.

10. Test your intuition, and look for counterexamples. Intuition suggests what to try, but only a proof or a calculation settles it. When a claim seems obvious, try to break it.

- *In this course:* disjoint events are not independent; pairwise independence is not mutual independence; $\mathrm{E}\, g(X) \neq g(\mathrm{E}X)$; a pdf can exceed 1; moments alone may not determine a distribution; the three prisoners.
- *Beyond:* correlation is not causation; one example does not prove a rule, but one counterexample disproves it.

11. Check your answer, preferably two ways. A wrong answer often fails a simple check.

- *In this course:* probabilities lie in $[0, 1]$; pmfs sum, and pdfs integrate, to 1; $\mathrm{Var}\, X \geq 0$; $M_X(0) = 1$; compute $\mathrm{E}\, g(X)$ from f_X and from $f_{g(X)}$; try a special case such as $n = 1$.
- *Beyond:* estimate the order of magnitude before you compute, and compare the result with your estimate.

Part IV. Before You Hand In

A short checklist for every problem.

- Did I define my events or random variables in words?
- Did I state supports and give cdfs and pdfs for **every** real x?
- Does every probability lie in $[0, 1]$? Does my pmf sum, or my pdf integrate, to 1?
- Did I cite the definition, axiom, or theorem behind each step?
- For an mgf, did I say where it is finite? Is $M(0) = 1$?
- Is $\mathrm{Var}\, X \geq 0$? Does a quick special case (e.g., $n = 1$, $p = \frac{1}{2}$) give a sensible answer?

Common mistakes.

- Using a Venn diagram as a proof of set equality.
- Treating disjoint events as independent, or assuming independence that is not given.
- Confusing $P(A \mid B)$ with $P(B \mid A)$.
- Counting the numerator and denominator in different ways (ordered vs. unordered).
- Forgetting the support, or the pieces of a cdf outside the support.
- Writing $\mathrm{E}\, g(X) = g(\mathrm{E}X)$ for nonlinear g.
- Computing $\mathrm{E}X$ for a distribution whose mean does not exist.
- Forgetting the absolute value $\left|\frac{d}{dy} g^{-1}(y)\right|$ for a decreasing g.

How to study. Rework Homeworks 1–3 and the Week 1–5 discussion problems without looking at the solutions, then redo the "Check your understanding" questions in the notes. For more practice, use the C&B exercises listed at the end of each week's notes (skip the §2.4 ones: 2.39 and 2.40).